

COSA+ and Dynamic Regime Adaptation

- Research question: can adaptation stay useful under sudden time-series shifts?
- Compare COSA+ online correction with Dynamic zero-shot adaptation.
- Main claim: trade-offs, not a universal winner.

Motivation

- Deployed streams shift due to sensor faults, weather anomalies, and demand spikes.
- Clean train/test splits hide this failure mode.
- We evaluate both clean performance and abrupt-shift recovery.

Baseline: COSA

- Frozen DLinear backbone.
- Online output-space adapter uses recent context.
- Lightweight, but still requires test-time gradient updates.

Our Two Directions

- COSA+: mean+std context and horizon-wise vector gate.
- Dynamic Regime Adapter: RegimeEncoder z plus GateGenerator g .
- COSA+ adapts online; Dynamic adapts by one forward pass.

Experimental Design

- Datasets: Weather and ETTh1.
- Backbone: frozen DLinear; horizons 96 and 720.
- Black-swan shifts: level, variance, trend, spike; metric: mse_after_50.

Main Results

- Weather h=96: Dynamic mixed loss has best clean MSE (+2.48%).
- ETTh1 h=720: COSA+ has best clean MSE (+4.86%).
- Under ETTh1 h=720 shifts, COSA+ gives strongest post-shift recovery.

Takeaways

- COSA+ helps when long-horizon errors accumulate after abrupt shifts.
- Dynamic Regime Adapter removes test-time backpropagation.
- Future work: gate regularization, shift detection, robust losses, more datasets.